KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Year 9 5.2 Yearly Exam: Help Sheet

Key ideas, formulas and worked examples for every skill in the practice exams

How to use it: keep this sheet beside you while you do a practice exam. When you mark your work, the answers page tells you which topic to read (for example **D4** or **M6**) for any question you got wrong. Read that card, then try the question again. Remember: in the real exam each question gives you the formula you need as a "Note".

Section 1: Data

D1 Types of data

- **Categorical**: words or groups, e.g. hair colour, favourite sport, shirt size.
- **Numerical discrete**: *counted*, so whole numbers, e.g. number of pets, goals, people.
- **Numerical continuous**: *measured*, so it can be any value, e.g. height, mass, time, temperature, volume.

Tip: ask "do I count it or measure it?"

D2 Shape of a distribution

- **Symmetrical**: both sides look the same; most scores in the middle.
- **Positively skewed**: scores bunched at the *low* end, tail stretching to the *right*.
- **Negatively skewed**: scores bunched at the *high* end, tail stretching to the *left*.
- **Cluster**: where most of the scores are grouped.

Tip: the skew is named after the side the tail is on.

D3 Reading graphs

- **Dot plot and histogram**: each column's height is the *frequency* (how many). The **mode** is the score under the tallest column, not the height of that column.
- **Total**: add up the heights of all the columns.
- **Percentage** = $\frac{\text{part}}{\text{total}} \times 100\%$

> **Example:** columns 5, 9, 4, 2. Total = 20. The column of 5 as a percentage: 5 ÷ 20 × 100% = **25%**.

D4 Mean, median, mode and range

> Mean = sum of the scores ÷ number of scores
> Range = highest score − lowest score

- **Median**: the middle score when the scores are *in order*. With an even number of scores, average the two middle ones.
- **Mode**: the score that occurs most often.

> **Example:** 4, 7, 7, 9, 13. Mean = 40 ÷ 5 = **8**; median = **7** (3rd score); mode = **7**; range = 13 − 4 = **9**.

D5 Outliers

An **outlier** is a score far away from all the others.

- It changes the **mean** the most (the mean uses every score).
- The median and mode hardly change, so the **median** describes data with an outlier better.

> **Example:** 6, 7, 7, 8, 32. Mean = 12 but median = 7. Without 32: mean 7, median 7.

D6 Stem-and-leaf plots: comparing two data sets

- Read each score with the key: **4 | 6 = 46**.
- In a **back-to-back** plot the left side is read backwards: a left leaf of 3 on stem 5 is 53.
- Which group scores higher? Compare the **medians** (or means).
- Which is **more consistent**? The one with the **smaller range**: its scores are closer together.

Always give the numbers in your reason: "Team A, because its range is 12, smaller than Team B's 30."

Section 2: Surface area and volume

M1 Units of capacity and volume

1 kL = 1000 L 1 L = 1000 mL
1 cm^3 = 1 mL 1 L = 1000 cm^3 1 m^3 = 1000 L

- Big unit → small unit: **× 1000**. Small unit → big unit: **÷ 1000**.

Examples: 3.2 kL = 3200 L; 850 mL = 0.85 L; 24 000 cm^3 = 24 L.

M2 Area of shapes

Shape	Area
Rectangle	$A = lb$
Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Kite or rhombus	$A = \frac{1}{2}xy$ (x, y = diagonals)

Use the **perpendicular height** (the dashed line at right angles), not the sloping side.

Example: triangle, base 10 cm, height 7 cm: ½ × 10 × 7 = **35 cm^2**.

M3 Circles

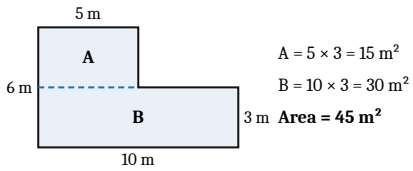
Circumference: $C = 2\pi r$ Area: $A = \pi r^2$

- The **radius** r is half the diameter. If the line goes right across the circle, halve it first.
- Use the π key and round only at the end.

Example: diameter 12 cm, so $r = 6$. $C = 2 \times \pi \times 6$ = **37.7 cm**; $A = \pi \times 6^2$ = **113.1 cm^2**.

M4 Composite shapes (L-shapes)

- **Perimeter**: add every outside side. Find missing sides first (opposite sides add up).
- **Area**: split into two rectangles and add, or take a big rectangle minus the missing corner.
- **Cost** = area × price per m^2.



M5 Volume

Rectangular prism: $V = lwh$ Cylinder: $V = \pi r^2 h$
Any prism: V = area of the end × length

Example: box 40 cm × 20 cm × 15 cm: V = 12 000 cm^3 = **12 L**.
Cylinder r = 3 cm, h = 10 cm: $V = \pi \times 3^2 \times 10$ = **282.7 cm^3**.

M6 Surface area (closed solids)

- **Box**: 6 faces in 3 matching pairs.
  $SA = 2lw + 2lh + 2wh$
- **Cylinder**: 2 circles + the curved rectangle.
  $SA = 2\pi r^2 + 2\pi rh$

Example: box 10 × 4 × 3 cm: 2(40) + 2(30) + 2(12) = 80 + 60 + 24 = **164 cm^2**.

Don't forget to double each face: there are 6 faces, not 3.

Section 3: Mixed Topics

A1 Substitution

Replace each letter with its number, then use BODMAS. **Powers come before ×.**

Example: if $a = 3$ and $b = 4$, then $2a^2 + b = 2 \times 3^2 + 4 = 2 \times 9 + 4$ = **22** (not $6^2 + 4$).

A2 Simplifying

- Only **like terms** (same letters) can be added or subtracted: $5a + 2b - 3a + b$ = **$2a + 3b$**.
- Multiply: numbers × numbers, letters × letters: $3a \times 4b$ = **$12ab$**.
- Divide: cancel numbers and letters: $20xy \div 4x$ = **$5y$**.

A3 Expanding brackets

Multiply every term inside by the term outside. A **negative** outside changes every sign inside.

Example: $3(2x + 1) - 2(x - 4)$
$= 6x + 3 - 2x + 8$ = **$4x + 11$**

$-2 \times -4 = +8$.

A4 Factorising

Take out the **highest common factor** (HCF) and put the rest in brackets. Check by expanding.

Examples: $8a + 12$ = **$4(2a + 3)$**
$m^2 - 6m$ = **$m(m - 6)$**

A5 Index laws

$a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$ $\quad (a^m)^n = a^{mn}$

Example: $3y^4 \times 5y^2 = 15y^6$ (multiply the numbers, **add** the indices).

E1 Solving equations

1. Expand any brackets.
2. Collect the pronumerals on one side (take away the smaller one).
3. Undo + and −, then × and ÷. Do the same to both sides.

Example: $5x - 2(x + 3) = x + 8$
$5x - 2x - 6 = x + 8 \rightarrow 3x - 6 = x + 8$
$2x = 14 \rightarrow \mathbf{x = 7}$. Check: 35 − 20 = 15 ✓

E2 Formulas and word problems

In $C = 40 + 25h$, 40 is the fixed fee and 25 is the cost per hour.

- **Find C**: substitute. For $h = 3$: $C = 40 + 75 =$ **$115**.
- **Given C**: write an equation and solve. $40 + 25h = 190 \rightarrow 25h = 150 \rightarrow \mathbf{h = 6}$.

F1 Percentages, discount and profit

Discount = % × marked price; sale price = marked price − discount
Profit = selling price − cost price
$\%\ \text{profit} = \frac{\text{profit}}{\text{cost price}} \times 100\%$

Example: \$80 less 15%: discount \$12, sale price **\$68**.
Bought \$200, sold \$260: profit \$60; 60 ÷ 200 × 100% = **30%**.

F2 Simple interest

$I = Prn$ (P = amount invested, r = rate as a decimal, n = years)

Example: \$3000 at 5% p.a. for 4 years: $I = 3000 \times 0.05 \times 4 =$ **\$600**.

Change the % to a decimal: 3.5% = 0.035.

F3 Earning money

1 year = 52 weeks = 26 fortnights = 12 months

- **Salary**: weekly = ÷ 52, fortnightly = ÷ 26, monthly = ÷ 12.
- **Overtime**: time-and-a-half = rate × 1.5 (double time = rate × 2). The first 38 hours are at the normal rate.
- **Commission** = % × sales. With a **retainer**, add the fixed amount.

Example: \$30/h, 41 hours: 38 × 30 = \$1140; 3 × 30 × 1.5 = \$135; total **\$1275**.

T1 Pythagoras' theorem

$c^2 = a^2 + b^2$ (c = hypotenuse, the longest side, opposite the right angle)

- Hypotenuse: **add** the squares, then √.
- Shorter side (e.g. a ladder's height up a wall): **subtract** the squares, then √.

Example: ladder 5 m, foot 1.5 m from the wall: $h^2 = 5^2 - 1.5^2 = 22.75$, so $h =$ **4.77 m**.

T2 Trigonometry: finding an angle

$\sin\theta = \frac{\text{opp}}{\text{hyp}}$ $\quad \cos\theta = \frac{\text{adj}}{\text{hyp}}$ $\quad \tan\theta = \frac{\text{opp}}{\text{adj}}$ (SOH CAH TOA)

1. Label the sides from the angle: opposite, adjacent, hypotenuse.
2. Pick the ratio that uses the two sides you know.
3. Use the inverse key: **SHIFT cos**.

Example: adj 1.5 m, hyp 5 m: $\cos\theta = 1.5 \div 5 \rightarrow \theta = \cos^{-1}(0.3) =$ **73°**.

Check the calculator shows D (degrees).